

GROVE, OK, USA

WMO: 722092

Lat: 36.605N

Lon: 94.738W

Elev: 254

StdP: 98.31

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53941

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-7.7	-16.1	0.9	-9.2	-12.7	1.3	-6.0	11.3	14.3	10.5	14.2	2.0	0	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	37.4	24.1	35.8	24.7	33.9	24.6	27.1	32.5	26.2	31.9	25.5	31.4	3.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	22.0	29.8	24.9	20.6	28.9	23.8	19.3	28.7	86.9	32.2	82.8	31.7	79.7	32.1	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.3	8.3	7.4	DB	-15.5	38.4	4.6	2.4	-18.8	40.2	-21.5	41.6	-24.1	42.9	-27.4	44.7	
			WB	-15.5	27.8	4.7	1.1	-18.9	28.6	-21.6	29.2	-24.2	29.8	-27.6	30.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	3.4	5.1	10.9	15.4	20.2	25.5	27.4	26.6	22.6	15.9	10.0	4.7
	DBStd	9.72	6.04	6.36	6.04	4.69	4.40	2.88	2.83	3.24	3.68	5.09	5.33	5.33
	HDD10.0	699	218	161	61	9	1	0	0	0	0	8	63	178
	HDD18.3	2012	464	371	243	109	33	1	0	0	7	108	253	422
	CDD10.0	2779	12	24	88	172	316	464	539	514	378	193	64	15
	CDD18.3	1049	1	1	11	23	90	214	281	256	135	34	4	1
	CDH23.3	11228	1	8	78	216	811	2289	3312	2868	1297	317	32	1
	CDH26.7	5114	0	1	9	38	248	1003	1724	1470	533	84	3	0
(14) Wind	WSAvg	2.5	2.8	2.9	3.2	3.4	2.6	2.4	1.8	1.7	1.8	2.2	2.7	2.6
(15) Precipitation	PrecAvg	1156	61	56	91	118	150	126	97	79	115	95	92	75
	PrecMax	1596	173	147	238	258	271	274	243	155	286	246	227	309
	PrecMin	775	7	8	21	32	49	35	15	7	17	31	8	10
	PrecStd	215	40	38	48	57	58	62	51	35	59	52	68	60
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	24.1	27.7	31.0	32.6	37.0	39.9	40.3	36.0	32.2	27.2	22.2
		MCWB	15.0	16.9	19.6	19.9	22.7	24.7	23.1	24.0	23.2	22.5	18.3	17.2
	2%	DB	19.0	22.1	26.0	27.5	31.3	34.9	37.5	37.5	32.9	28.9	23.7	19.1
		MCWB	12.8	15.0	18.1	18.2	23.0	25.3	24.0	24.2	23.3	21.0	16.6	14.1
	5%	DB	17.0	18.7	22.9	26.2	29.8	32.9	35.9	35.9	32.2	27.2	21.3	16.3
		MCWB	11.0	14.2	16.7	17.9	22.1	24.6	24.7	24.7	23.0	19.1	16.2	11.9
	10%	DB	13.6	16.2	21.2	23.9	27.6	32.2	34.0	33.6	30.2	24.8	19.2	13.7
		MCWB	9.5	11.1	15.0	17.2	21.1	24.4	25.1	24.1	22.4	18.8	14.5	10.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.9	18.3	21.8	21.8	25.6	28.4	28.8	27.6	25.6	23.7	20.3	17.9
		MCDB	20.8	22.5	26.2	27.1	31.1	33.7	33.8	33.4	31.6	29.0	24.1	20.6
	2%	WB	14.3	16.5	19.6	20.3	24.1	26.8	27.1	26.6	24.7	22.6	18.6	15.9
		MCDB	17.0	20.5	23.9	25.3	28.9	32.1	32.3	32.6	30.7	27.2	22.0	18.1
	5%	WB	11.9	14.5	17.9	19.4	23.1	25.7	26.3	25.8	23.9	21.4	17.0	12.9
		MCDB	15.7	18.1	22.5	24.2	27.2	31.2	31.8	31.9	29.4	24.8	20.4	15.2
	10%	WB	9.4	11.6	15.9	18.3	22.1	25.0	25.6	25.1	23.2	20.0	15.0	10.8
		MCDB	14.0	15.2	20.1	22.6	26.5	30.4	31.4	31.3	28.2	23.2	18.4	13.0
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.4	12.2	12.2	11.1	10.9	11.7	11.6	12.2	12.4	12.4	11.0
		MCDBR	16.3	15.4	14.6	14.0	12.7	12.0	13.7	14.0	13.8	14.4	14.5	14.8
		MCWBR	10.4	10.1	8.4	7.5	6.0	4.7	4.0	4.2	5.2	7.0	9.4	10.0
		MCDBR	13.5	13.7	12.1	11.4	10.9	10.3	10.9	11.4	11.7	10.4	12.2	12.1
	5% WB	MCWBR	10.2	9.9	7.7	7.3	6.0	5.2	4.9	4.9	5.1	6.0	9.2	9.8
(40) Clear-Sky Solar Irradiance	taub		0.305	0.311	0.350	0.385	0.432	0.430	0.448	0.451	0.394	0.350	0.315	0.294
	taud		2.465	2.462	2.390	2.309	2.236	2.269	2.264	2.234	2.358	2.462	2.505	2.515
	Ebn at Noon		890	929	916	897	856	854	835	825	858	872	872	882
	Edh at Noon		83	94	111	128	140	135	135	136	114	94	80	75
(44) All-Sky Solar Radiation	RadAvg		2.45	3.02	4.02	5.08	5.65	6.49	6.46	5.83	4.95	3.65	2.63	2.08
	RadStd		0.22	0.45	0.24	0.30	0.48	0.56	0.40	0.40	0.38	0.46	0.29	0.26

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page